# Homework

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• regression • classification

## 1 Goal

I am giving you a bank marketing dataset. Each row is one phone call to a client, and the last column, `y`, says whether that client subscribed to a term deposit (yes or no). Your job is to predict `y` from the other columns.

The data is `train.csv` in the folder: https://drive.google.com/drive/u/0/folders/1e4PoI5xp-aTN6M5PH5wiSwX6QkBblayS

Here is a guide to the data too: `feature-guide.txt` (same folder).

## 2 Ground rule

Please don't look this dataset up online, and ask your AI not to either. The whole point is to build it yourself and understand it.

## 3 Part 1: logistic regression (individual)

- This one is on your own.
- The math is the easy part and AI will help. You have to do the preprocessing.
- The data is full of text columns and "unknown" values, and a model only reads numbers. So decide, column by column, how to turn the text into numbers and how to handle the unknowns and anything else messy.
- Design that yourself. Ask AI to help you think it through if you want, but the decisions are yours.
- Then run a plain logistic regression.
- Report all of your scores: accuracy, precision, recall, F1, and ROC-AUC.
- Share your AI logs and conversations.
- Use AI to help with your code, but write the code yourself (recommended).

## 4 Part 2: your own problem (teams)

- This part is by teams.
- Find an interesting machine learning problem, not this one.
- Solve it with one model of your choice.
- Do some interesting and creative feature engineering.
- Understand how your model actually works, and be ready to explain it.
- Bring its scores too.
- Share your AI logs and conversations.

## 5 Why bother

- Part 1 is a good learning experience.
- Part 2 is a good project showcase.

- I did something similar my freshman year with my friends, a plant-disease classifier, and put it on my resume: https://github.com/Sam-B-Y/CV-Corn-Disease-Detection. You can and should do this too.

## 6 Note

More details on how to submit your final model are coming soon.